

Computer assignment 2

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```
set.seed(031026)
load("proj_data.Rdata")
modell1 <- glm(Resultat ~ Alder + Kon + Utbildare,
               data = data_individ,
               family = "binomial")
summary(modell1)

##
## Call:
## glm(formula = Resultat ~ Alder + Kon + Utbildare, family = "binomial",
##      data = data_individ)
##
## Coefficients:
##              Estimate Std. Error z value Pr(>|z|)
## (Intercept)   -0.065131    0.253358  -0.257  0.79712
## Alder         -0.025122    0.008671  -2.897  0.00376 **
## KonMan         0.100567    0.137049   0.734  0.46307
## UtbildareTrafikskola 0.947498    0.158405   5.982 2.21e-09 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## (Dispersion parameter for binomial family taken to be 1)
##
##      Null deviance: 1344.4  on 999  degrees of freedom
## Residual deviance: 1291.4  on 996  degrees of freedom
## AIC: 1299.4
##
## Number of Fisher Scoring iterations: 4

## Functions
## test1
# -----a-----
L <- function(theta,y,X){
  z <- as.vector(X %*% theta)
  p <- 1/(1+exp(-z))
  prod(dbinom(y, size=1, prob=p))
}

l <- function(theta,y,X){
  z <- as.vector(X %*% theta)
  # p <- plogis(z)
  p <- 1/(1+exp(-z))
  sum(dbinom(y, size=1, prob=p, log=TRUE))
}#using 'dbinom' in both L and l
```

```

S <- function(theta,y,X){
  z <- as.vector(X %*% theta)
  p <- 1/(1+exp(-z))
  t(X) %*% (y-p)
}

I <- function(theta,y,X){
  z <- as.vector(X %*% theta)
  p <- 1/(1+exp(-z))
  D <- diag(as.vector(p*(1-p)))
  t(X) %*% D %*% X
}

# -----b-----
NR <- function(theta0, niter, y, X){
  for (i in 1:niter) {
    score <- S(theta0, y, X)
    fisher <- I(theta0, y, X)
    theta0 <- theta0 + solve(fisher, score)
  }
  return(theta0)
}

y <- matrix(data_individ$Resultat, ncol = 1)
X <- model.matrix(Resultat ~ Alder + Kon + Utbildare,
                  data = data_individ)

```

Task 1

```

# Coefficient of glm method
theta_mo <- coef(model1)
theta_mo <- as.numeric(theta_mo)

# Use functions I and NR from Part I
theta0 <- c(0,0,0,0)
max_iter <- 100
for (i in 1:max_iter){
  theta0 <- NR(theta0, niter=1, y, X)
  theta_hat <- theta0
  if (all(round(theta_hat, 2) == round(theta_mo, 2))) {
    cat("niter =", i, "reaches two-digits accuracy")
    break
  }
}

## niter = 2 reaches two-digits accuracy

# Calculate standard error and Wald statistic
fisher_info <- I(theta_hat, y, X)

```

```

fisher_inv <- solve(fisher_info)
std_errors <- sqrt(diag(fisher_inv))
wald_stats <- theta_hat/std_errors
print(wald_stats)

##              [,1]
## (Intercept) -0.2602174
## Alder       -2.8925231
## KonMan      0.7331498
## UtbildareTrafikskola 5.9803387

# Compare the z value column of the output
r_z_values <- summary(model1)$coefficients[, "z value"]
#print(r_z_values)

Difference <- round(abs(wald_stats-r_z_values), 6)
print(Difference)

##              [,1]
## (Intercept)  0.003145
## Alder       0.004721
## KonMan      0.000657
## UtbildareTrafikskola 0.001167

```

The differences between `wald_stats` and `r_z_values` being less than 0.01 confirms that the z-values in R's summary output are indeed Wald statistics.

Task 2

```

# Under H0
eta <- NR(theta0 = c(0,0,0),niter=10, y=y, X=X[, -3])
eta

##              [,1]
## (Intercept)  0.00215153
## Alder       -0.02524899
## UtbildareTrafikskola 0.92960275

L_eta <- L(eta,y,X=X[, -3])

# Without H0 restriction
L_raw <- L(theta_hat,y,X)

# Calculate generalized likelihood ratio
ratio <- L_raw/L_eta
W <- 2*log(ratio)
cat("generalized likelihood ratio statistics:", W, "\n")

## generalized likelihood ratio statistics: 0.539802

```

```
# p-value
p_value <- pchisq(W, df = 1, lower.tail = FALSE)
cat("P-values:", p_value, "\n")

## P-values: 0.4625148
```

Since the Wald statistic follows an asymptotic standard normal distribution, its square follows a chi-square distribution with one degree of freedom. The generalized likelihood ratio statistic also asymptotically follows a chi-square distribution, and in this problem its parameter (q) is one. Therefore, both statistics are of the same order of magnitude.

Task 3

```
# Fit the model under  $H_0: \vartheta = (\vartheta_{\text{intercept}}, \vartheta_{\text{Kon}}) = (0, 0)$ 
X_H0 <- X[, c(2, 4)]

# Estimate  $\eta$  under  $H_0$ 
eta_H0 <- NR(theta = c(0, 0), niter = 10, y = y, X = X_H0)

# Construct full parameter vector
theta0 <- c(0, 0) # (intercept, Kon) = (0, 0)
beta_H0 <- c(theta0[1], eta_H0[1], theta0[2], eta_H0[2])

# Compute the score vector
score_H0 <- S(beta_H0, y, X)

# Compute Fisher information matrix
info_H0 <- I(beta_H0, y, X)

# Compute the generalized score statistic
#  $T_S = S(\vartheta_0, \eta_{ML})^T * I(\vartheta_0, \eta_{ML})^{-1} * S(\vartheta_0, \eta_{ML})$ 
T_S <- t(score_H0) %*% solve(info_H0) %*% score_H0
T_S <- as.numeric(T_S)
cat("Generalized Score Statistic T_S:", T_S, "\n")

## Generalized Score Statistic T_S: 0.5387157

# Compute p-value (asymptotically  $\chi^2(2)$ )
p_value <- 1 - pchisq(T_S, df = 2)
cat("P-value for  $H_0$ :", p_value, "\n")

## P-value for  $H_0$ : 0.7638699
```

Task 4

```
# Compute wald interval
# Z-crit
z_crit <- qnorm(0.975)
```

```

# Wald interval
Wald_L <- theta_hat[3] - z_crit * std_errors[3] # Lower Confidence Limit
Wald_U <- theta_hat[3] + z_crit * std_errors[3] # Upper Confidence Limit
cat("Wald interval of theta_kon is:", Wald_L, "to", Wald_U)

## Wald interval of theta_kon is: -0.1681245 to 0.3690684

# # Compute profile likelihood at a fixed theta_Kon=t:
# function
Profile_likelihood <- function(t){
  theta.Kon <- t
  profile <- glm.fit(x= X[, -3], y = y,
                    offset = theta.Kon * X[,3],
                    family = binomial())
  pro_coeff<- profile$coeff
  # plug back
  linear_pred <- profile$linear.predictors
  probs <- binomial()$linkinv(linear_pred)
  log_lik <- sum(dbinom(y, size = 1, prob = probs, log = TRUE))
  pro_lik <- exp(log_lik)
  return (pro_lik)
}

# test
LP_thetakon <- Profile_likelihood(0.5)
LP_thetakon

## [1] 5.74353e-283

# Compute Estimated Likelihood
eta_global <- coef(model1)[-3]
Estimated_likelihood <- function(t){
  theta.Kon <- t
  linear_pred <- X[, -3] %*% eta_global + X[,3] * theta.Kon
  probs <- binomial()$linkinv(linear_pred)
  log_lik <- sum(dbinom(y, size = 1, prob = probs, log = TRUE))
  pro_lik <- exp(log_lik)
  return (pro_lik)
}

# test
Le_thetakon <- Estimated_likelihood(0.5)
Le_thetakon

## [1] 6.240748e-286

# Compute Profile and Estimated value
theta_range <- seq(-0.5,0.5,by=0.001)

```

```

LP_thetakon <- sapply(theta_range, Profile_likelihood)
Le_thetakon <- sapply(theta_range, Estimated_likelihood)

# Find 95% interval
LP_max <- max(LP_thetakon)
Pro_relative <- LP_thetakon / LP_max
Re_p <- qchisq(0.95,df=1)
threshold <- exp(-Re_p /2)
LP_CI95 <- threshold * LP_max
LP_CI95

## [1] 5.569285e-282

## Draw
# y_range
y_lim_all <- range(c(LP_thetakon, Le_thetakon))

# Profile Likelihood
plot(theta_range, LP_thetakon,
     type = "l",
     lwd = 3,
     col = "blue",
     xlab = expression(theta[Kon]),
     ylab = "Likelihood Value",
     main = "Profile vs Estimated Likelihood Function",
     ylim = c(min(y_lim_all), max(y_lim_all)),
     las = 1
)

# Estimated Likelihood
lines(theta_range, Le_thetakon,
     lwd = 3,
     col = "red",
     lty = 2)

# 95% confidence interval
abline(h = LP_CI95,
     col = "darkgreen",
     lty = 3,
     lwd = 2)

# Wald interval
abline(v = c(Wald_L, Wald_U),
     col = "orange",
     lty = 3,
     lwd = 2)

# Background & Label
grid(col = "gray", lty = 1, lwd = 0.5)
legend("topright",

```

```

legend = c("Profile",
           "Estimated",
           "95% threshold",
           "Wald interval"),
col = c("blue", "red", "darkgreen", "orange"),
lwd = c(3, 3, 2, 2),
lty = c(1, 2, 3, 3),
bty = "n"
)

```

Profile vs Estimated Likelihood Function

